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Mental Health Prediction Using Human Emotions with Swin Transformer

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I. ABSTRACT

Facial Expression Recognition (FER) has become a key part of affective computing. Facial cues offer dependable clues about a person's emotions and frame of mind. Given the increase in mental health concerns all over the world, methods for keeping track in a way that doesn't disrupt daily life have become more vital, especially in the workplace, healthcare, and research. This work puts emphasis on using vision models, in particular the Swin Transformer, to help spot mental health states by looking at facial emotions. These models use datasets like FER2013 and CK+, which allow the system to spot basic emotions such as happiness, sadness, anger, fear, surprise, and a neutral state.

This project's goal is to create emotion-recognition systems that can precisely spot early signs of student stress, depression, and other risks to mental health in real-time. Current FER methods, namely those making use of CNNs, typically struggle with slight changes in facial expressions, perform strangely in uncontrolled settings, and don't explain how certain parts of the face lead to how emotions are predicted. This project also mentions problems, such as datasets that are not balanced, limited ability to adapt, and a disconnect between recognizing emotions and scoring mental health in a way that is helpful.

To fix this, the system uses Swin Transformer models improved with attention and interpretation tools like Grad-CAM. The design includes a way to score mental health by using average confidence and the number of images looked at. By mixing feature extraction based on transformers, organized data preprocessing, and evaluation scripts, the repository gives a framework for emotion-based mental health studies without going into how the ideas are put into action.

INDEX TERMS: Facial Emotion Recognition (FER), Mental Health Prediction, Swin Transformer, Temporal Modeling.

II. INTRODUCTION

Background:

Mental-health disorders such as depression and chronic stress have emerged as major global challenges, impacting academic performance among students, workplace productivity, and overall well-being. Early psychological disturbances often manifest through subtle changes in facial behaviour, making facial expressions an important non-verbal channel for understanding internal affective states. With the increasing interest in unobtrusive and automated mental-health monitoring systems, facial- expression analysis has become a promising approach for identifying early indicators of psychological distress.

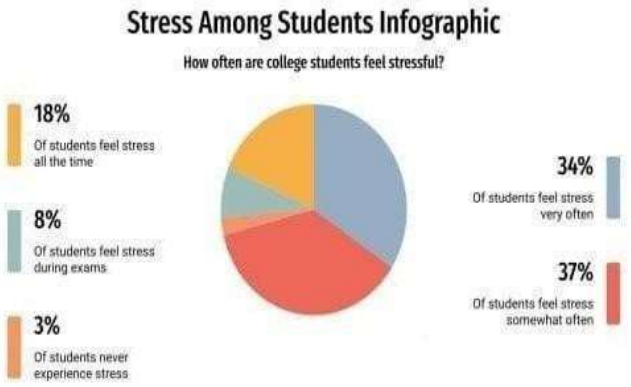

Figure [a]

Figure [a] *presents how often students feel stressed, with data on different stress levels among them*

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The Need / Motivation :

Standard ways of checking mental health, like interviews and questionnaires, rely on what people say, are skewed, and do not keep track of things over time. People might not share how they feel because they are embarrassed or unsure. Therefore, we need tools that are objective, always on, and don't bother people. These tools should spot actions linked to mental health risks.

Figure [b] *presents stats that show common school- related stress factors, such as not getting enough sleep, pressure from schoolwork and society, and overall stress among college students.*

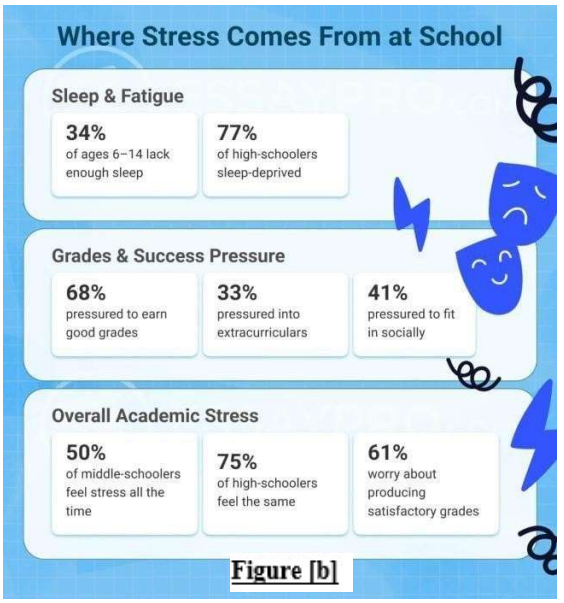

Figure [b]

Existing Work:

Recent studies from 2022 to 2025 show quick growth in Facial Emotion Recognition (FER) that is powered by transformer-based models. Swin Transformers, types of CNN-Transformer models, and Action Unit (AU)-based systems have gotten ahead of CNNs in understanding face movement, micro-expressions, and context. Some studies take FER a step further into mental health by looking at AU actions, how unsteady emotions are, emotion patterns over time, and quick bursts of micro-expressions.

Drawbacks:

Even with these steps, papers point out problems that limit how accurately we can predict mental health. Current datasets don't have mental health info and have unequal classes, unreliable labels, biases, and not enough real-world differences. Most models look at pictures only, ignoring emotion patterns over time, which are key for judging mental health.

Proposed Approach:

The idea is to create a system that is easy to use and can guess a person's mental health by looking at their facial expressions. The system uses the Swin Transformer model because it spots small and big facial details better than past models. First, the system finds face images or video frames, lines them up, and adjusts them to get rid of errors. Then, the Swin Transformer pulls out facial features like eye movements, eyebrow position, mouth shape, and overall expression.

To improve the prediction, the design has a function that keeps track of emotions over time. This looks at how someone's face changes instead of just one view. It helps see how stable someone is and is key to guessing about their mental health. Last, the design offers a scoring module to show likely stress, worry, or depression levels.

III. NEED OF THE STUDY

Mental health has become a major concern in today's fast-paced and stressful lifestyle, especially among students and working individuals. Early identification of emotional instability is essential to prevent severe mental health issues such as anxiety, depression, and stress-related disorders. However, existing mental health assessment methods mainly rely on self-reporting, questionnaires, or clinical consultations, which may not always provide accurate or timely results. Many individuals hesitate to openly express their feelings, leading to delayed diagnosis and treatment.
Additionally, traditional systems lack real-time monitoring and continuous analysis of emotional patterns. This creates a gap in understanding a person's day-to-day mental condition. To address these challenges, there is a need for an automated, non-invasive, and efficient system that can detect emotions directly from facial expressions. The proposed system fulfills this need by using deep learning techniques to analyze emotions and predict mental health status, enabling early intervention and better emotional well-being management.

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IV. FUNDAMENTAL OF EMOTIONS AND DEEP LEARNING

Basics of Emotions:

Emotions are core feelings that show how people respond to situations, experiences, and what's inside them. They affect how people act, think, decide, and feel overall. Basic human emotions that are usually found in psychology are happiness, sadness, anger, fear, disgust, surprise, and feeling neutral. These emotions can be seen through face movements like raising eyebrows, changing lip corners, tightening eyes, and changing mouth shape. Because facial expressions happen naturally and without us trying, they show how someone feels at the moment. Repeated emotion patterns can point to mental health states, like long-term stress, worry, or sadness.

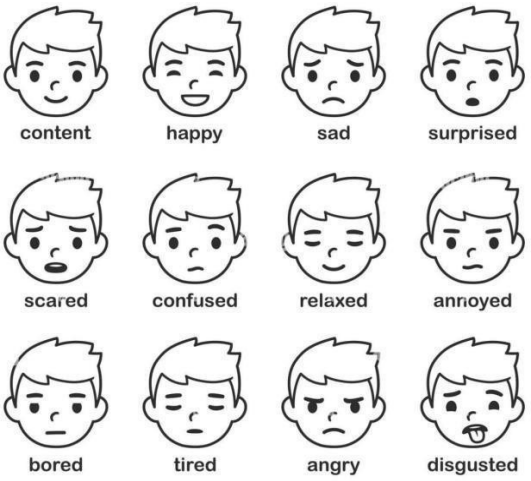

Figure [c]

Figure [c] *presents simple icons for twelve basic emotions that are used when studying facial expression.*

Basics of Deep Learning:

Deep learning is an element of artificial intelligence that trains multi-layer neural networks to independently pull out data from raw input like images, sound, or words. Unlike traditional machine- learning methods that need hand-made data extraction, deep-learning models pick up patterns straight from big datasets. This makes them good at jobs such as sorting images, face matching, and emotion research. Convolutional Neural Networks (CNNs) were key in image-related tasks for a time, but Transformer-based designs - like Vision Transformer (ViT) and Swin Transformer - have grown. These models apply attention to both local details and overall relationships in an image, which allows for face understanding.

Overview of Emotions and Deep-Learning Concepts:

Emotions help in understanding our behaviors and state of mind. Facial expressions can give a look into the emotional. Deep learning allows for automatic analysis of face data without hand design. By reviewing large amounts of images, deep-learning models can recognize muscle changes linked to emotion like sadness, anger, fear, or joy. Modern designs like Swin Transformer make this better by using focus approaches to capture face characteristics and relationships. By aligning emotion recognition and deep-learning methods, an emotional detection framework can be set that has a way of guessing at potential mental health risks through behavior patterns.

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V. LITERATURE REVIEW

S.No	Year	Title	Method Used	Measures	Key Outcomes	Limitations
1	2025 [1]	Real-Time Classroom Emotion Analysis Using Machine and Deep Learning for Enhanced Student Learning	Hybrid ML + Deep Learning (CNN, SVM, Ensemble models)	Accuracy, real-time performance	Real-time classroom emotion detection improves teaching insights and engagement	Requires high-quality video input; model complexity
2	2024 [2]	Mental Health and Stress Prediction Using NLP and Transformer-Based Techniques	Transformer-based NLP (BERT, RoBERTa)	Accuracy, precision, recall	Transformer models effectively predict stress from text patterns	Limited to text input; contextual ambiguity
3	2023 [3]	Vision Transformer With Attentive Pooling for Robust Facial Expression Recognition	Vision Transformer (ViT) + Attentive Pooling	Accuracy, robustness	Attentive pooling significantly enhances ViT performance for FER	Requires large datasets; computationally heavy
4	2021 [4]	SwinTransformer: Hierarchical Vision Transformer using Shifted Windows	Swin Transformer (Shifted Windows)	Accuracy, speed, scalability	Hierarchical design improves efficiency and accuracy in vision tasks	High computational demand for training
5	2023 [5]	Facial Expression Recognition Algorithm Based on Swin Transformer	Swin Transformer for FER	Accuracy, F1 score	Swin-based FER improves feature extraction and recognition accuracy	Dataset limitations; high model complexity
6	2024 [6]	Swin-FER: Swin Transformer for Facial Expression Recognition	Swin Transformer + optimized FER pipeline	Accuracy, robustness	Achieves high FER accuracy with improved attention modeling	Limited generalization across demographic variations
7	2023 [7]	Facial Expression Recognition using Squeeze and Excitation-powered Swin Transformers	Swin Transformer + SE blocks	Accuracy, feature enhancement	SE blocks further enhance expressive feature extraction	Relatively high computational cost
8	2025 [8]	Mental health assessment model for college students based on facial expression recognition."	CNN + FER-based psychological assessment model	Accuracy, Evaluation metrics	FER-based mental health assessment improves early detection	FER-only; lacks physiological signals
9	2024 [9]	Emotion Recognition of the Elderly Based on Transformer and Physiological Signals	Transformer + physiological features (HRV, ECG, GSR)	Accuracy, sensitivity	Multi-signal fusion improves emotion recognition for elderly users	Elderly-specific dataset; not generalizable

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S.No	Year	Title	Method Used	Measures	Key Outcomes	Limitations
10	2025 [10]	Emotional recognition and intervention of students in intelligent English teaching environment based on deep learning	Deep Learning (CNN, RNN, attention models)	Accuracy, intervention effectiveness	Deep learning improves emotion detection in smart English classrooms	Requires continuous monitoring; privacy concerns
11	2025 [11]	Emotion recognition for enhanced learning: using AI to detect students' emotions and adjust teaching methods	AI + FER + adaptive teaching models	Accuracy, adaptability, usability	AI-based emotion detection supports personalized learning	Privacy, need for high-quality video data
12	2025 [12]	Deep learning based emotion recognition for analyzing students' psychological states during competitions	Deep Learning (CNN, Hybrid models)	Accuracy, robustness	Helps assess psychological states during high-pressure events	Limited to competition settings
13	2023 [13]	Real-Time Attention Monitoring System for Classroom: A Deep Learning Approach for Student's Behavior Recognition	Deep Learning-based behavior recognition	Accuracy, response time	Effective real-time monitoring of student attention	Hardware-dependent; requires clear visuals
14	2022 [14]	Real-time mental stress detection using multimodality expressions with a deep learning framework	Multimodal DL (face + voice + physiological signals)	Accuracy, F1 score	High-accuracy stress detection using multimodal fusion	Requires synchronized multimodal inputs

IV. PROPOSED SYSTEM

The proposed system focuses on predicting mental health by analyzing human facial emotions using a deep learning-based approach. It utilizes a Swin Transformer model to accurately detect emotions from facial images or real-time video input. The system is designed to be non-invasive, efficient, and capable of providing continuous monitoring of a user's emotional state.

The architecture consists of several stages, including image input, preprocessing, emotion detection, and mental health score generation. Initially, the input image or video frame is processed to detect and extract facial features. The Swin Transformer model then classifies the detected face into one of the predefined emotion categories. These emotions are stored and aggregated over a specific time period.

The workflow involves capturing facial data for approximately 30 seconds, during which multiple emotion predictions are recorded. Based on the frequency and intensity of detected emotions, a mental health score is calculated. This score helps in identifying whether the user is emotionally stable or requires attention, enabling timely intervention.

V. METHODOLOGY

The methodology of the proposed system is divided into multiple stages to ensure accurate emotion detection and mental health prediction. Data collection is performed using the FER2013 dataset, which contains labeled facial images representing different human emotions. This dataset provides a reliable foundation for training the deep learning model. In the data preprocessing stage, images are resized to a standard dimension, converted to grayscale, and normalized to improve model performance and reduce computational complexity. These steps help in extracting meaningful features while minimizing noise.

The model architecture is based on the Swin Transformer, which efficiently captures both local and global features from facial images using attention mechanisms. For emotion detection, the model classifies images into seven categories: happy, sad, angry, fear, surprise, disgust, and neutral. Finally, mental health scoring is performed by analyzing detected emotions over time. A baseline score (e.g., 50) is adjusted based on positive and negative emotions, providing an overall indication of the user's mental state.

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VI. IMPLEMENTATION

The implementation of the proposed system is carried out using Python as the primary programming language due to its simplicity and extensive support for machine learning libraries. The deep learning model is developed using PyTorch, which provides flexibility for building and training neural networks. OpenCV is used for image processing tasks such as capturing facial images, detecting faces, and preprocessing input data in real time.
The training process involves feeding preprocessed facial images from the FER2013 dataset into the Swin Transformer model. The model is trained over multiple epochs to learn patterns associated with different emotions. The Cross-Entropy loss function is used to measure the difference between predicted and actual labels, while the AdamW optimizer is applied to update the model weights efficiently. During each training iteration, the model performs forward propagation, loss calculation, and backpropagation to improve accuracy. This process enables the system to reliably classify emotions and support mental health prediction.

VII. CHALLENGES

A. *Dataset Limitations and Imbalanced Emotion Classes*

Recent papers bring up datasets that have irregular emotion distributions, with only a few signs of disgust and fear. This inequality can shift learning. Models tend to overfit to the predominant emotions and misclassify rare ones.

B. *Real-World Variations Reduce Accuracy*

Studies from 2022–2025 note that FER may be changed by lighting, camera quality, background, contact (masks, glasses), head positions, and hair. Lack of response can make things hard when putting systems to use when checking on mental health in daily settings.

C. *Difficulty in Capturing Micro- Expressions*

Micro-expressions last for only a fraction of a second and reveal emotional states that individuals may attempt to hide. Literature shows that even transformer-based models have difficulty detecting these subtle cues due to insufficient micro-expression datasets and the need for extremely high temporal resolution

D. *Lack of Reliable Mental-Health Labels*

With current models and emotions, the mental health section includes datasets for stress, depression, and related issues. Studies depend on questionnaires, which are inconsistent and subjective.

E. *Identity, Age, and Ethnicity Bias*

Research explains that traits are changed by age, identity, and race. Rather than reviewing patterns, transformer models may take the patterns they see and utilize them regardless of the subject. Some demographics may be incorrect because of this, bringing on unfairness and complications with mental health.

F. *Overfitting in Transformer Models*

Transformers have big models and need various amounts of datasets. Since most of the datasets are not the biggest, models may be fit to training samples. Without change, transformer structures can be used to memorize face patterns rather than learning patterns.

G. *Ethical and Privacy Concerns*

During 2022 and 2025, it's been spoken about that there are ethical considerations for face-based mental- health prediction. Storing data, predicting mental conditions, and potential use of data raise ethical concerns. Without regulations, execution is hard, despite technological advancements.

VIII. RESULTS AND DISCUSSION

The proposed system was evaluated based on its ability to accurately detect facial emotions and generate a meaningful mental health score. The Swin Transformer model demonstrated good performance in classifying the seven emotions, achieving reliable accuracy on the FER2013 dataset. The model was able to effectively distinguish between different emotional states, even with variations in facial expressions and lighting conditions.
During testing, it was observed that the system performs well in real-time scenarios, capturing facial data and producing consistent emotion predictions over the 30-second analysis period. The aggregation of emotions helped in generating a stable and interpretable mental health score. Sample outputs showed that positive emotions resulted in lower scores, while negative emotions increased the score, indicating potential emotional distress. Overall, the system provides accurate and consistent results, making it suitable for real-time mental health monitoring and preliminary emotional assessment applications.

IX. FUTURE SCOPE

Future work can be conducted within facial-expression-based mental-health prediction in directions that can be built upon. With continuing growth in architectures like Swin Transformers, focus can be given to efficient and lightweight models that may be put to use in real-time and low end devices. Mental-health checkup systems may be put to use on devices, workspace setups, and devices without sacrificing accuracy. Research can be put into emotion dataset expansions with representation from age groups, races, structures, and settings. Present datasets go through limited variety and inequality, which limits models. Improving datasets can make transformer-based FER better when conducting real time scenarios, studies have the option to go to data-changing methods and create a way to study emotion. With mental health being connected to emotions rather than expressions, checking sequential structures and periodic recordings may put focus into mood or depression. Work being conducted has areas such as scoring to find emotions and change them into indicators. Trend- based analysis, frequency maps, and more can make predictions sensible and clinical friendly. Visual based explanation and region-based improvements can increase the ethics and safety of the user. With responsible AI being significant, future AI models can have increased measures to make health check up's safe.

X. CONCLUSION

Facial-expression-based mental-health prediction is a point where computer vision and analysis come together. Enhancements in deep learning and transformer structure like the Swin Transformer has enhanced the power of small emotional indicators and patterns. This provides a foundation to developing models that can analysis emotions and find signs of possible stress,depression and health problems among the students.

Literature from 2022 and 2025 shows growth in model correctness, safety, and comprehension. Limits consist of dataset diversity, the need for structure, and psychological scoring. Works show that transformer- based FER approaches show clear advantages in handling emotional expressions.

Findings suggest steps can be taken forwards to create a Swin-Transformer-based mental-health predictor. By balancing datasets, data input, emotion mapping, and visualization, system and ethical support can be built for safe and reasonable use. With transformer-driven FER containing growth and assistance for early assessment, research and applications can be built for future exploration.

XI. ACKNOWLEDGEMENT

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